

GEOTHERMAL DEVELOPMENT COMPANY

MONITORING AND EVALUATION SYSTEM DESIGN

Paka-Baringo Silali 100MW Geothermal Steam Field Development

A Results-Based Management Framework for a KfW and GRMF
Funded Geothermal Energy Programme in Kenya

PROJECT BUDGET USD 201.3M KfW + GRMF + GOK + AMEA	TARGET CAPACITY 70 – 100 MW Baseload by June 2030	M&E INDICATORS 19 Indicators Input → Output → Outcome → Impact
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PATHWAY 1 Energy Cost Reduction	PATHWAY 2 Climate and Emissions	PATHWAY 3 Energy Security	PATHWAY 4 Community Welfare
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DISCLAIMER: Data presented in this document is illustrative and based on simulated programme scenarios grounded in the GDC Project Charter Version 3.0 (June 2025) and EPRA FY2023/2024 statistics. It does not represent actual GDC operational data.

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1. Nature of Data Presented

All data, figures, indicator values, baseline measurements, targets, financial figures, monitoring results, community survey findings, and dashboard visualizations presented in this M&E System Design document are illustrative and based on simulated programme scenarios. They do not represent actual operational, financial, or confidential data held by the Geothermal Development Company.

The simulated data has been constructed specifically to demonstrate the architecture, logic, and functionality of the M&E system including realistic variance, milestone progression, and assumption status changes. The dataset covers a 24-month illustrative window from January 2024 to December 2025, representing Year 1 and Year 2 of the steam field development phase of the project. It is important to note that this 24-month window is not the full monitoring period the complete M&E system is designed to operate across the entire project lifecycle from 2025 through to 2032, spanning the steam field development phase (2025–2027), power plant construction phase (2027–2030), commissioning and early operations phase (2030–2031), and the post-commissioning impact evaluation phase (2031–2032). The simulated dataset illustrates how the monitoring system would function during the steam field development phase only. It is designed to show how the system would operate in practice, not to report on actual project performance at any phase of the project lifecycle.

2. Baseline and Reference Data Sources

Baseline figures used in this document come from two different kinds of publicly available sources, plus a set of illustrative figures as outlined below.

Regulatory and government data

Source	Data Used
EPRA Bi-Annual Energy and Petroleum Statistics Report, FY 2023/2024	Kenya installed geothermal capacity (943.7MW), geothermal share of generation mix (41.71%), national grid emission factor (0.5 tCO2/MWh), total installed capacity (3,243MW)
EPRA Monthly Gazette Notices, November 2025	Fuel Energy Cost Charge (FECC) reference of KES 3.81 per kWh
Kenya LCPDP 2022 to 2041	Power demand growth projections (8 to 10%), geothermal capacity target (5,000MW by 2030), grid installed capacity data
IPCC and EPRA Methodologies	CO2 emissions avoided calculation: MW x capacity factor x 8,760 hours x 0.5 tCO2/MWh grid emission factor

GDC's own public communications

Source	Data Used
GDC official website, gdc.co.ke/baringo	KfW EUR 80 million loan and its funding purpose, Government of Kenya funding of access roads and well pads, 300MW first phase target across Paka, Korosi, and Silali at 100MW each, ESIA licence from NEMA

GDC 2025 Information Booklet	20 exploration wells completed at Paka, 85MW of steam realised at Paka
GDC blog, CleanEnergyWeek post	AMEA Power named as GDC's partner for Paka field development
Public procurement tender, tenders.go.ke	Confirms “Strategic Partner” as the official term for the IPP role and the 70MW to 100MW capacity range
Kenya Investment Promotion Authority, opportunities.invest.go.ke	Public project listing for the Baringo-Silali-Paka geothermal project
Public statements by GDC leadership and government officials, press coverage 2023 to 2025	GDC's 1,065MW by 2030 company target, indicative tariff range of 6 to 7 US cents per kWh, approximate well cost of USD 5 million

Illustrative figures

Source	Data Used
No external source, illustrative	Project budget breakdown (USD 201.3M by component), KfW loan terms including interest rate and grace period, internal planning assumptions including cost per well and cost per MW, staffing structure, and component level completion percentages

3. Purpose and Status of This Document

The purpose of the document is to demonstrate professional competency in results-based management, monitoring and evaluation system design, indicator development, data collection methodology, evaluation planning, and donor accountability frameworks.

Eric Kizambo, Head of Corporate Performance Management Section at the Geothermal Development Company, designed this M&E system for the Paka-Baringo Silali project, a live geothermal development that GDC is currently implementing. The Theory of Change, indicator framework, monitoring tools, and evaluation plan reflect the same M&E practice applied throughout this document, designed against a real project, not a hypothetical one.

The specific figures shown in the tables and dashboard are illustrative. GDC's actual reporting data is internal, so the numbers here stand in for it to show how the system functions in practice.

Project context referenced throughout, including budget figures, financing terms, and technical assumptions, is illustrative and reflects internal project planning information. Any use of this document beyond personal portfolio purposes should be confirmed with GDC first.

4. What This Document Does Not Represent

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- Actual GDC operational performance data. All indicator values, monitoring results, and dashboard figures are illustrative.
- Actual KfW, GRMF, or Government of Kenya disbursement records. Financial figures are illustrative.
- Actual well drilling results, steam capacity measurements, or reservoir engineering assessments. Technical figures are illustrative planning estimates, not field measurements.

- Actual community survey findings. Survey results are simulated based on plausible programme trajectories.
- Actual AMEA Power commercial agreements, PPA terms, or power plant construction status. Commercial milestone status is illustrative.
- An official or approved GDC M&E framework. This document has not been reviewed or endorsed by GDC management, KfW, or GRMF.
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5. Intellectual Property

The M&E system design, results framework architecture, Theory of Change, indicator framework, data collection tools, evaluation plan structure, monitoring matrix design, and dashboard visualization presented in this document are original professional work. The analytical frameworks, methodological approaches, and system design choices reflect the author's professional judgement and expertise.

Project context, including GDC institutional information, project budget, financing structure, and project assumptions, is illustrative and reflects internal project planning information. It is reproduced here in summary form for personal portfolio purposes. Broader use or distribution should be confirmed with GDC.

6. Data Management Standards Applied

The simulated dataset and monitoring dashboard have been constructed in accordance with World Bank DIME Data Handbook reproducibility and transparency standards. Specifically:

- All data is version controlled. The master dataset is identified as GDC_PBS_ME_Simulated_Dataset_v1.0.xlsx.
- Data sources are documented for every baseline figure drawn from a public source. Figures not drawn from a public source are clearly marked as illustrative.
- Calculation methodologies are transparently documented, particularly the CO2 emissions avoided calculation using the EPRA grid emission factor.
- The disclaimer is prominently displayed on the dashboard, cover page, and document header, ensuring no reader can mistake illustrative figures for actual performance data

SUMMARY DISCLAIMER STATEMENT

Data presented in this M&E System Design document is illustrative, based on simulated programme scenarios, and grounded where possible in publicly available EPRA FY2023/2024 statistics and GDC's own public communications about the Paka-Baringo Silali project. It does not represent actual GDC operational, financial, or technical data. The 24 month illustrative dataset represents Year 1 and Year 2 of the steam field development phase only; the full M&E system is designed to operate across the entire project lifecycle from 2025 to 2032. The M&E system design, frameworks, methodologies, and analytical approaches are original professional work, designed for a live GDC project. This document has not been formally reviewed or endorsed by GDC, KfW, or GRMF.

Eric Kizambo | Head, Corporate Performance Management Section | GDC | December 2025

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1. Executive Summary

1.1 Project Overview

The Paka-Baringo Silali 100MW Geothermal Steam Field Development is one of GDC's flagship geothermal projects and a nationally strategic investment financed by KfW (EUR 80 million), the Geothermal Risk Mitigation Facility (GRMF), and the Government of Kenya, with AMEA Power as the private Strategic Partner responsible for constructing and operating the power plant under a 25-year Public-Private Partnership agreement.

GDC's mandate is to develop geothermal steam through exploration, drilling, and steam field infrastructure thereby removing the upfront resource risk that deters private sector investment in geothermal power generation. The Paka Sub-Project targets 70 to 100MW of geothermal electricity capacity by 2030, contributing directly to Kenya's Vision 2030 energy objectives, the Least Cost Power Development Plan target of 5,000MW of geothermal by 2030, and Kenya's Nationally Determined Contribution under the Paris Agreement.

By month 24 of the illustrative dataset, GDC has drilled 34 wells, confirmed 22 productive, and established 100MW of steam capacity at wellhead, the project's critical steam field development milestone. The Steam Supply Agreement with AMEA Power is complete, the Power Purchase Agreement between AMEA Power and KPLC is under EPRA review, and the Steam Gathering System is 52 percent complete. These figures are simulated for purposes of this M&E system design, see the Disclaimer for full detail.

1.2 Purpose of This M&E System

This M&E system is designed to serve four functions simultaneously:

- Accountability to KfW and GRMF as financing partners
- Operational management decision-making for GDC project teams
- Learning to support replication of GDC's PPP model in future geothermal fields
- Transparency to communities and the Kenyan public

The framework draws on internationally recognized M&E standards, including OECD DAC evaluation criteria and DIME data management practices, adapted to GDC's own reporting needs.

1.3 Theory of Change — Four Causal Pathways

- **Pathway 1: Energy cost reduction.** Geothermal electricity generated at a projected cost of 6.5 US cents per kWh displaces expensive thermal and emergency power costing 16 to 22 US cents per kWh, thereby contributing to downward pressure on electricity generation costs for Kenyan households and businesses.
- **Pathway 2: Climate and emissions reduction.** Every MWh of geothermal electricity generated displaces fossil fuel thermal generation thereby avoiding an estimated 277,000 to 394,200 tonnes of CO₂ equivalent annually at 70 to 100MW capacity, contributing to Kenya's Nationally Determined Contribution (NDC) target of a 32 percent emissions reduction by 2030 under the Paris Agreement.
- **Pathway 3: Energy security.** Adding 70 to 100MW of reliable geothermal baseload power and operating at 90 to 95 percent capacity factor regardless of weather, diversifies Kenya's generation mix currently at 22.8 percent hydro and 17.7 percent thermal, reducing dependence on sources vulnerable to drought and fuel price volatility.
- **Pathway 4: Community welfare.** Infrastructure development, including 70km of paved roads and community water supply systems serving 7 treatment plants and 14 cattle troughs, combined with local employment creation, delivers direct and sustained welfare benefits to communities in the Baringo-Silali project area.

The Theory of Change is supported by eight monitored assumptions, covering well productivity, IPP investment, PPA viability, KETRACO transmission delivery, steam quality, and regulatory approvals, each tracked monthly through the M&E system.

1.4 Results Framework and Key Indicators

The M&E system tracks 19 indicators across five results levels, from inputs through to impact, set out in full in the Results Framework and Monitoring Matrix (Section 4) and the Indicator Reference Sheet (Section 5). The table below highlights ten of the most decision-relevant indicators.

Level	Indicator	Baseline	Final Target
Input	Budget utilised (IN.1)	63%	100%
Output	Steam capacity at wellhead (OP.6)	85 MW	100 MW
Output	Wells drilled cumulative (OP.4)	20 wells	34 wells
Output	SGS construction (OP.7)	0%	100%
Imm. Outcome	Steam supply agreement (IO.1)	Not signed	Signed, achieved
Imm. Outcome	PPA ratified by EPRA (IO.2)	Not ratified	Ratified
Int. Outcome	MW commissioned (MO.1)	0 MW	70–100 MW
Impact	CO2 avoided annually (IM.2)	0 tCO2e	394,200 tCO2e
Impact	Electricity generation cost (IM.1)	22¢/kWh emergency	6.5¢/kWh
Impact	Geothermal capacity added (IM.3)	0 MW from Paka	70–100 MW

1.5 M&E Implementation: Data Collection, Evaluation, and Reporting

The M&E Plan (Section 6) operationalizes this framework through a four-stage data flow, from field and administrative data generation, through monthly aggregation and a four-step quality assurance process, to reporting via six report types: monthly dashboard updates, monthly internal reports, quarterly donor progress reports, bi-annual community engagement reports, annual results reports, and ad hoc issue reports. An estimated annual M&E budget of USD 264,000 covers staffing, community surveys, environmental monitoring, and data systems.

Four data collection tools (Section 7) generate the underlying evidence: a bi-annual Community Perception and Welfare Survey measuring awareness, water access, employment, and grievances across a minimum 200 households; a monthly Environmental and Social Safeguards Checklist covering eight compliance domains; a monthly Project Performance Tracking Template covering drilling, infrastructure, financial, and commercial milestone progress; and a bi-annual Key Informant Interview Guide capturing outcome progress and stakeholder perspectives from AMEA Power, EPRA, KfW, KPLC, and community leaders.

The Evaluation Plan (Section 8) applies the six OECD DAC criteria across three evaluations spanning the project’s full lifecycle: a mid-term review (USD 40,000 to 60,000), a pre-commissioning evaluation (USD 80,000 to 120,000), and a post-commissioning impact evaluation using contribution analysis and difference-in-differences methodology (USD 120,000 to 180,000).

Progress against all of the above is visualized in the M&E Monitoring Dashboard (Section 9), presented across two pages for print legibility.

1.6 Illustrative Project Status and Next Steps

The table below summarizes milestone status at the month 24 of the illustrative dataset.

Milestone	Status
Access roads, 70km paved	Achieved
Water supply system	84% complete, on track
Drill pads, 21 commissioned	Achieved
Wells drilled, 34 cumulative	Achieved
Wells confirmed productive, 22	Achieved
Steam capacity, 100MW at wellhead	Achieved
Steam supply agreement, GDC / AMEA Power	Achieved
SGS construction, 52% complete	On track
PPA ratification by EPRA	In progress, under review
Power plant commissioning	Pending, later project phase

The single item outstanding at this stage is PPA ratification by EPRA, the critical enabling condition for AMEA Power’s Financial Investment Decision and the start of power plant construction. Once ratified, the M&E system’s focus shifts from steam field development indicators toward the intermediate outcome and impact indicators set out in Section 4.

2. Programme Context and Stakeholder Analysis

2.1 GDC Institutional Background and Mandate

The Geothermal Development Company Limited is a wholly government-owned parastatal established under the laws of Kenya with a specific and exclusive mandate: to accelerate the development of geothermal energy resources across Kenya's Rift Valley system. GDC's operational model is deliberately designed to address the primary market failure that constrains private sector participation in geothermal development: the high cost and technical uncertainty of upstream resource exploration and development.

GDC absorbs the upfront exploration, appraisal, and production drilling risks using concessional financing from development partners, removing the resource risk that deters private capital from investing in geothermal power generation. Once the resource is proven and steam confirmed at commercially viable quantities, GDC makes it available to private sector Independent Power Producers through steam supply agreements, enabling IPPs to invest in power plant construction and operation without bearing the upstream geological risk.

GDC's mandate aligns directly with Kenya's Vision 2030 energy objectives and the Least Cost Power Development Plan 2022 to 2041, which projects power demand growth of 8 percent annually, projected to exceed 10 percent upon full Vision 2030 implementation. GDC is specifically tasked with developing at least 1,065MW of geothermal capacity by 2030. The Paka Sub-Project is the most advanced of GDC's pipeline projects outside the established Olkaria field and represents a critical test of GDC's PPP development model in a new geothermal zone.

2.2 Baringo-Silali Geothermal Block: Resource and Location

The Baringo-Silali Geothermal Block is a significant unexploited geothermal resource zone, a large volcanic complex situated in Baringo County in the northern Kenyan Rift Valley. The block comprises three sub-projects, Paka, Korosi-Chepchuk, and Silali, with a combined estimated resource potential of approximately 3,000MWe.

The Paka geothermal field is situated 25 kilometers north of Lake Baringo. The Paka volcanic center rises to a maximum altitude of 1,697 meters above sea level, covering approximately 280 square kilometers. Surface geothermal manifestations e.g. hot springs, fumaroles, and altered ground cover approximately 45 square kilometers, with reservoir temperatures confirmed above 270 degrees Celsius, among the highest readings in Kenya's geothermal fields. Detailed surface exploration estimates a resource potential of 500MWe at Paka alone, providing significant headroom above the current 70 to 100MW project target.

GDC commenced drilling at Paka in December 2018 and has drilled 20 wells, confirming the existence of a viable geothermal resource with over 85MW equivalent of steam available at wellhead from the wells drilled to date. A total of 34 wells are required to support the 70 to 100MW power plant target.

2.3 Kenya's Energy Context: The Development Problem

Kenya's electricity sector is characterized by four structural vulnerabilities that the project directly addresses:

- Weather-dependent generation dominance: Hydropower output is highly variable, dependent on rainfall patterns increasingly disrupted by climate change and El Niño variability. During drought periods, hydro generation falls sharply, forcing Kenya to procure expensive emergency thermal power.
- Expensive emergency and thermal generation: Kenya procures emergency power at approximately 22 US cents per kilowatt hour and oil-based thermal generation at approximately 16 US cents per kilowatt hour. The Paka project is projected to deliver power at approximately 6.5 US cents per kilowatt hour, a 70 percent reduction against emergency power costs.

- Insufficient domestic baseload capacity: Kenya's total installed capacity as of June 2024 was 3,243MW, of which 943.7MW is geothermal (24.69% of installed capacity). The LCPDP projects demand growth requiring significant new generation capacity additions, with geothermal identified as the preferred baseload technology given its 90 to 95 percent capacity factor.
- Private sector deterred by upstream geothermal risk: Geothermal development requires substantial upfront investment in exploration and drilling before the resource is confirmed, a risk that private investors are typically unwilling to bear. GDC's model of absorbing this upstream risk through concessional development finance creates the conditions for private sector power plant investment.

2.4 Project Description

The Paka Sub-Project aims to develop 70 to 100MW of geothermal generation capacity within the Baringo-Silali Block by facilitating and establishing the preconditions for complete steam field development and the generation and commercialization of geothermal electricity by a private Strategic Partner. The project is implemented under a PPP framework in which GDC develops the steam field and makes steam available, while AMEA Power, the engaged Strategic Partner, finances, constructs, and operates the power plant for 25 years.

The project commenced in July 2010. Infrastructure development began in 2015 with major components commissioned by 2018, including 70 kilometers of paved access roads. Exploratory drilling commenced in December 2018 and has since confirmed geothermal resources with 85MW equivalent of steam confirmed at wellhead from 20 wells drilled. The project is expected to reach power plant commissioning before the end of the decade.

The total project budget is USD 201.3 million, financed through a KfW concessional loan of EUR 80 million (signed July 2014, effective November 2016), GRMF grant financing, Government of Kenya budgetary allocations, and AMEA Power's private investment in the power plant and steam gathering system.

2.5 Policy and SDG Alignment

Policy Framework	Project Alignment
Kenya Vision 2030	Explicitly identified as a Vision 2030 energy project contributing to Kenya's economic transformation energy pillar.
LCPDP 2022–2041	Contributes directly to LCPDP geothermal generation targets of 5,000MW by 2030. Geothermal identified as preferred baseload technology.
Kenya National Energy Policy 2025–2034	Supports target of 100 percent renewable electricity generation by 2030 through expanding geothermal baseload capacity.
Kenya NDC — Paris Agreement	Projected 277,000 to 394,200 tCO ₂ e avoided annually contributes to Kenya's 32 percent emissions reduction commitment by 2030.
SDG 7 — Affordable Clean Energy	Directly contributes to SDG 7.1.1 (electricity access), 7.2.1 (renewable energy share), 7.3.1 (energy intensity), 7.b.1 (installed renewable capacity).
SDG 13 — Climate Action	Emissions avoided directly contribute to SDG 13.2.2 (total greenhouse gas emissions per year).
SDG 8, 6, 9 and 17	Local employment (SDG 8.5), community water access (SDG 6.1), resilient infrastructure (SDG 9.1), and PPP framework (SDG 17.17) are direct project co-benefits.

2.6 Stakeholder Analysis

The following matrix maps all primary and secondary stakeholders, their roles, interests, influence levels, and M&E information needs.

Primary Stakeholders

GDC — Geothermal Development Company	
Role	Project owner and implementer, responsible for steam field development, well drilling, infrastructure construction, and steam supply to AMEA Power.
Interest	Successful steam field development enabling steam supply revenue; proof of PPP model for replication; mandate delivery toward 1,065MW by 2030.
Influence	Very High
M&E data needs	Full results chain monitoring, OP.1 through IM.3; assumption monitoring; ESCP compliance.
Reporting frequency	Monthly operational; quarterly progress; annual results

AMEA Power — Strategic Partner	
Role	Finances and constructs the steam gathering system (Build-Transfer) and 70–100MW power plant; operates plant for 25 years under PPP.
Interest	Commercially viable PPA tariff; reliable steam supply at agreed quality and pressure; stable regulatory environment.
Influence	Very High
M&E data needs	Steam capacity confirmation (OP.6); PPA ratification (IO.2); generation performance (MO.2, MO.3); emissions data.
Reporting frequency	Quarterly during construction; monthly once operational

KfW — German Development Bank	
Role	Primary development finance partner, EUR 80M concessional loan financing water supply, drilling, consultancies, feasibility, and transaction advisory.
Interest	Loan utilisation against disbursement schedule; project delivery against results framework; ESCP compliance; climate impact evidence.
Influence	Very High
M&E data needs	Full results chain, IN.1 through IM.3; disbursement-linked milestones; environmental compliance; CO2 avoided data.
Reporting frequency	Quarterly progress; annual results; supervision missions

GRMF — Geothermal Risk Mitigation Facility	
Role	Grant financing for geothermal risk mitigation, managed by AfDB, funded by EU.
Interest	Demonstration that Paka resource risk is successfully mitigated, enabling replication of risk mitigation model across Africa.
Influence	High
M&E data needs	Resource development milestones (OP.5, OP.6); climate impact evidence (IM.2); replication evidence.
Reporting frequency	Semi-annual progress reports

Ministry of Energy and Petroleum	
Role	Policy and regulatory oversight, issues geothermal license to GDC; coordinates all sector players.
Interest	Project contributes to national energy targets; demonstrates government commitment to geothermal development.
Influence	High
M&E data needs	Generation capacity additions (IM.3); energy cost competitiveness (IM.1); NDC contribution (IM.2).
Reporting frequency	Annual sector report

EPRA — Energy and Petroleum Regulatory Authority	
Role	Independent energy regulator, ratifies PPA; issues generation license to AMEA Power; sets and reviews tariffs.
Interest	Commercially viable and consumer-protective PPA tariff; generation license compliance; grid reliability.
Influence	Very High
M&E data needs	PPA ratification milestone (IO.2); generation performance (MO.2, MO.3); capacity factor compliance.
Reporting frequency	Monthly generation performance reports

KPLC — Kenya Power and Lighting Company	
Role	National electricity off-taker, enters PPA with AMEA Power; enters steam compensation agreement with GDC.
Interest	Reliable, affordable electricity at competitive tariff; PPA terms enabling cost recovery; grid stability.
Influence	High
M&E data needs	Generation volume (MO.2); capacity factor (MO.3); electricity cost (IM.1).
Reporting frequency	Monthly offtake data

KETRACO — Kenya Electricity Transmission Company	
Role	Plans, designs, constructs, and commissions power transmission and evacuation infrastructure for Paka.
Interest	Timely project implementation enabling grid connection by April 2027.
Influence	High
M&E data needs	Transmission construction progress against project schedule, critical path assumption (AS.5).
Reporting frequency	Quarterly construction progress

Government of Kenya — Ministry of Finance	
Role	Project financier through budgetary allocations; sovereign guarantor for KfW loan.
Interest	Efficient use of public resources; project delivery against national energy targets; debt management.
Influence	High
M&E data needs	Budget utilisation (IN.1, IN.2); project delivery against KfW commitments; value for money evidence.
Reporting frequency	Annual budget performance reporting

Secondary Stakeholders

Stakeholder	Role	Interest	Influence	M&E Data Need
Local Communities	Direct beneficiaries and project-affected persons, pastoralist communities in Paka field area.	Water supply access; local employment; infrastructure improvements; transparent benefit-sharing.	Medium	Community welfare indicators; water access (OP.2); grievance resolution.
Independent Consultants	Implementation Consultant; Quality Management Consultant; Environmental and Social Management Consultant; Feasibility Study Consultant.	Contractual deliverable completion; professional standards compliance.	Medium	Varies by role, QMC uses OP.4 and OP.5; Environmental Consultant uses safeguards monitoring data.
GDC Staff	Project implementation team, drillers, geo-scientists, engineers, environmentalists, economists.	Successful project delivery; professional development; GDC revenue from steam supply.	Medium	Project progress against plan; resource development milestones.

The Kenyan Public	Ultimate beneficiaries, households and businesses benefiting from increased electricity supply and lower costs.	Affordable, reliable electricity; reduced power outages; lower electricity bills over time.	Low	Electricity cost trend (IM.1); grid reliability (MO.3); national renewable energy share (IM.3).
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3. Theory of Change

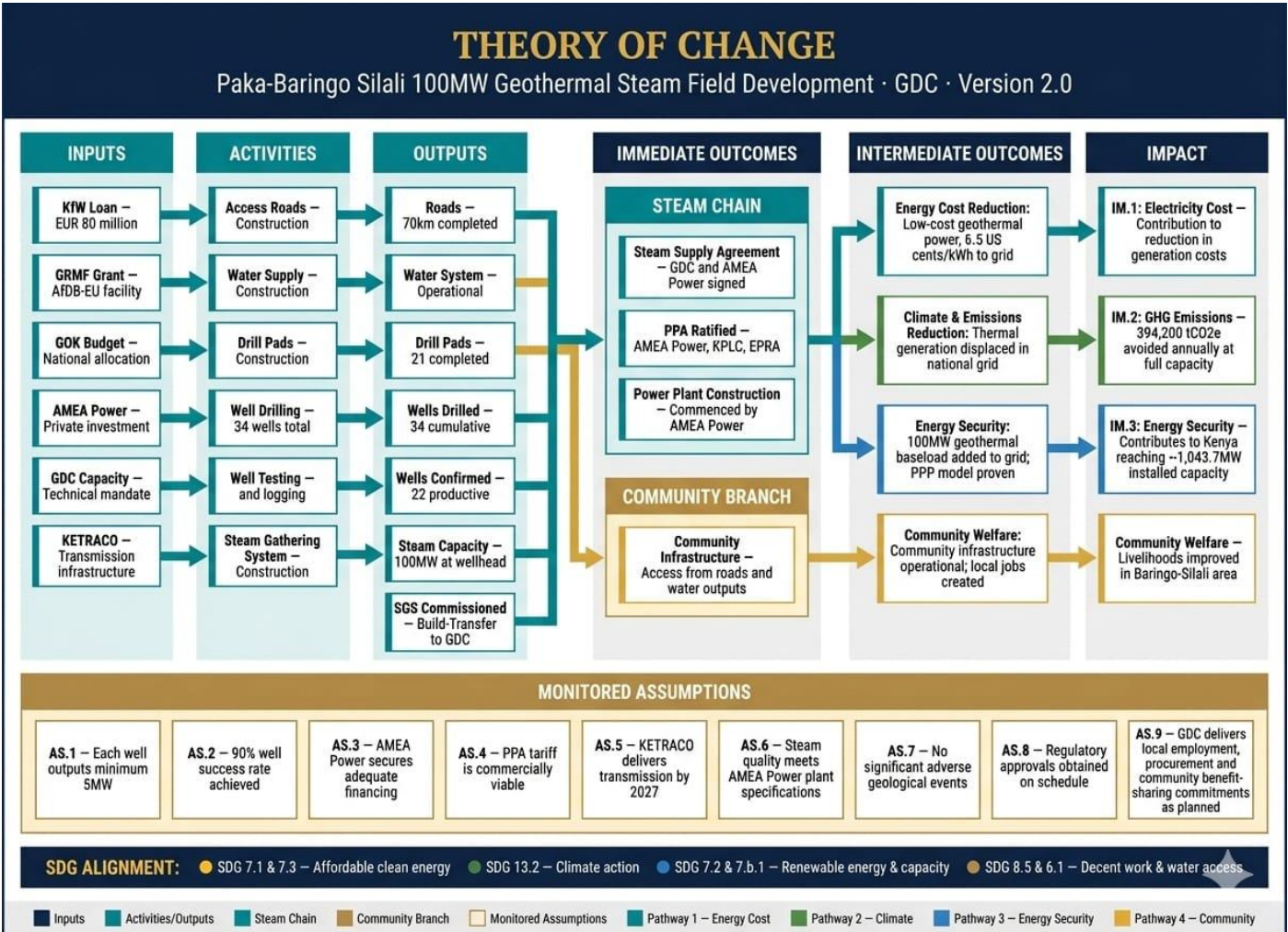
3.1 Problem Statement

Kenya faces a fundamental energy challenge that constrains economic growth, increases household and business energy costs, and threatens climate commitments. Power demand is growing at 8 percent annually, projected to exceed 10 percent upon full implementation of Vision 2030, while the generation mix remains dangerously dependent on weather-vulnerable hydropower and expensive imported fossil fuel thermal generation.

Emergency power currently costs 22 US cents per kWh and oil-based power costs approximately 16 US cents per kWh. Hydropower is increasingly unreliable due to climate-driven drought and El Niño variability. Geothermal energy offers the solution: abundant, indigenous, weather-independent, low-cost, and low-emission. GDC exists specifically to remove the market barrier of upfront geothermal resource risk.

3.2 Theory of Change Diagram

The diagram below presents the project’s Theory of Change at a strategic level, showing the causal logic from inputs through activities, outputs, immediate outcomes, intermediate outcomes, and impact, across four parallel causal pathways. Monitored assumptions are shown in line at each transition point in the results chain. Full indicator-level detail corresponding to each element of this diagram is provided in the Results Framework and Monitoring Matrix (Section 4) and the Indicator Reference Sheet (Section 5).



3.3 Four Causal Pathways: Narrative

Pathway 1: Energy Cost Reduction

GDC develops geothermal steam through drilling and testing of production wells in the Paka field. Confirmed steam capacity enables AMEA Power to invest in power plant construction under a commercially viable PPA with KPLC. The power plant generates electricity at a projected cost of 6.5 US cents per kWh, compared to 22 cents for emergency power and 16 cents for oil-based generation. As geothermal power enters the generation mix at this significantly lower cost, it displaces expensive thermal generation, reducing the fuel energy cost component of Kenya's electricity tariff over time.

Causal chain: Wells drilled and confirmed → Steam capacity available → AMEA Power plant commissioned → Low-cost electricity generated → Thermal generation displaced → Electricity generation costs reduced → Energy affordability improved.

Key assumptions: AS.4: PPA tariff agreed between AMEA Power and KPLC reflects actual generation cost and is commercially viable for sustained operation; AS.3: AMEA Power secures adequate financing for power plant construction.

Pathway 2: Climate and Emissions Reduction

Electricity generated from geothermal steam is inherently low emission. Every MWh generated by the Paka plant displaces an MWh that would otherwise have been generated from thermal sources, avoiding the associated greenhouse gas emissions calculated using Kenya's national grid emission factor of 0.5 tCO₂ per MWh published by EPRA. At full 100MW capacity operating at 90 percent capacity factor, the Paka plant will avoid approximately 394,200 tonnes of CO₂ equivalent annually.

Causal chain: Wells drilled and confirmed → Steam capacity available → AMEA Power plant commissioned → Geothermal electricity generated → Thermal generation displaced → CO₂ emissions avoided → Kenya's NDC commitments advanced.

Key assumption: AS.8: All regulatory approvals including the EPRA generation license are obtained on schedule, allowing the displacement pathway to materialize as planned.

Pathway 3: Energy Security and Diversification

Geothermal addresses Kenya's energy security vulnerabilities simultaneously: unlike hydro, geothermal is unaffected by rainfall variability, operating at 90 to 95 percent capacity factor regardless of climate conditions; unlike thermal, geothermal uses a domestic resource eliminating fuel import dependency. By adding 70 to 100MW of geothermal baseload capacity, the project contributes to diversification of the generation mix and demonstrates the PPP model for replication at Korosi-Chepchuk and Silali.

Causal chain: Wells drilled and confirmed → Steam capacity available → AMEA Power plant commissioned → Geothermal baseload added to grid → Generation mix diversified → Weather and import dependence reduced → Energy security strengthened.

Key assumption: AS.5: KETRACO delivers transmission infrastructure enabling power evacuation from Paka to the national grid by April 2027.

Pathway 4 — Community Welfare and Economic Development

Unlike Pathways 1 through 3, this pathway does not depend on the steam supply agreement, PPA ratification, or power plant construction. Community benefits from infrastructure begin as soon as roads and water systems are operational, regardless of AMEA Power's commercial progress. Infrastructure development, including 70km of paved roads and water reticulation systems serving communities through 7 treatment plants and 14 cattle troughs, provides lasting community assets that improve mobility, access to markets, and animal welfare in a pastoralist community. Employment creation during construction and operations phases provides income to local community members, generating economic multiplier effects in Baringo county.

Causal chain: Infrastructure constructed → Community assets and access improved → Local employment and procurement implemented → Household income increased → Community welfare improved.

Key assumption: GDC implements its local employment and procurement commitments and community benefit-sharing mechanisms are established and communicated transparently.

3.4 Assumptions: Explicitly Mapped to the Theory of Change

The Theory of Change holds only if the conditions below remain true throughout project implementation. Each assumption is mapped to the specific causal pathway it governs: an assumption failure at any level breaks the causal chain for the pathways that depend on it. Assumptions are not background notes; they are actively monitored through the assumptions monitoring section of the Monitoring Matrix (Section 4.6) and reported monthly through the Project Performance Tracking Template (Tool 3, Section 7.3). Where an assumption moves from Holding to At Risk, the M&E Plan’s adaptive management trigger points (Section 6.10) define the specific management response required.

Code	Assumption	Pathway Affected	Risk Level
AS.1	Each well outputs minimum 5MW	Pathways 1, 2, 3	Medium
AS.2	90% well success rate achieved	Pathways 1, 2, 3	Medium
AS.3	AMEA Power secures financing for power plant	Pathways 1, 2, 3	High
AS.4	PPA tariff is commercially viable for AMEA Power	Pathways 1, 2, 3	High
AS.5	KETRACO delivers transmission by April 2027	Pathways 1, 2, 3	Medium
AS.6	Steam quality meets AMEA Power plant specifications	Pathways 1, 2, 3	Low
AS.7	No significant adverse geological events	Pathways 1, 2, 3	Low
AS.8	Regulatory approvals obtained on schedule	Pathways 1, 2, 3	Low
AS.9	GDC delivers local employment, procurement, and community benefit-sharing commitments as planned	Pathway 4	Medium

3.5 Theory of Change: Monitoring Matrix Linkage

Every element of the Theory of Change is operationalized through a named indicator in the Results Framework and Monitoring Matrix (Section 4). This linkage is what distinguishes a Theory of Change from a conceptual diagram that stands alone: each box in the diagram above is tracked, measured, and reported through the monitoring system described in the remainder of this document.

Theory of Change Element	Monitoring Matrix Indicator(s)
Inputs — financing	IN.1, IN.2
Activities — staffing and technical capacity	IN.3
Outputs — infrastructure (roads, water, drill pads)	OP.1, OP.2, OP.3
Outputs — wells drilled, tested, and confirmed	OP.4, OP.5, OP.6
Outputs — steam gathering system	OP.7
Immediate outcomes — commercial agreements	IO.1, IO.2

Immediate outcomes — plant construction commenced	IO.3
Intermediate outcomes — generation and capacity	MO.1, MO.2, MO.3
Impact — Pathway 1 (Energy Cost)	IM.1
Impact — Pathway 2 (Climate and Emissions)	IM.2
Impact — Pathway 3 (Energy Security)	IM.3
Assumptions — all eight, inline on diagram	AS.1 through AS.8

Theory of Change Element	Monitoring Matrix Indicator(s)
Inputs: financing	IN.1, IN.2
Activities: staffing and technical capacity	IN.3
Outputs: infrastructure (roads, water, drill pads)	OP.1, OP.2, OP.3
Outputs: wells drilled, tested, and confirmed	OP.4, OP.5, OP.6
Outputs: steam gathering system	OP.7
Immediate outcomes: commercial agreements (steam chain)	IO.1, IO.2
Immediate outcomes: plant construction commenced (steam chain)	IO.3
Immediate outcome: community infrastructure access (Pathway 4 branch)	Evidenced by OP.1 and OP.2 completion; dedicated indicator to be confirmed, see implications document
Intermediate outcomes: generation and capacity	MO.1, MO.2, MO.3
Impact: Pathway 1 (Energy Cost)	IM.1
Impact: Pathway 2 (Climate and Emissions)	IM.2
Impact: Pathway 3 (Energy Security)	IM.3
Assumptions: all nine, inline on diagram	AS.1 through AS.9

3.6 What the Theory of Change Does Not Claim

This is as important as what it does claim. The project does not claim to:

- Directly reduce household electricity bills — tariff reduction depends on EPRA regulatory decisions and KPLC distribution cost structures outside GDC's control
- Solely determine Kenya's grid emission factor trajectory — this is influenced by all generation sources in the mix
- Directly electrify Baringo communities — grid connection depends on KPLC last-mile investment outside GDC's mandate
- Guarantee IPP power plant delivery — AMEA Power's investment decision depends on PPA viability, financing access, and regulatory approvals partially outside GDC's direct control

These limitations are acknowledged in the contribution versus attribution framing applied to each impact indicator in Section 5 — Indicator Reference Sheet.

4. Results Framework and Monitoring Matrix

This section operationalises the Theory of Change presented in Section 3 into a measurable results framework. Every indicator below traces directly back to a specific element of the Theory of Change — the linkage table in Section 3.5 provides the full cross-reference. The monitoring matrix tracks 19 result indicators across five levels — three input indicators, seven output indicators, three immediate outcome indicators, three intermediate outcome indicators, and three impact indicators — plus eight assumption monitoring items corresponding to the eight assumptions identified in Section 3.4. All indicators include baselines grounded in the GDC Project Charter Version 3.0 and EPRA FY2023/2024 published statistics. Impact-level indicators are framed using contribution rather than attribution language — acknowledging that GDC’s steam field development contributes to national energy outcomes alongside multiple other sector actors, consistent with the limitations set out in Section 3.6.

4.1 Input Indicators

Code	Indicator	Baseline	Yr 1Target	Yr 2Target	Data Source	Collection Method	Frequency	Responsible
IN.1	Project budget disbursed as % of USD 201.3M approved budget	63%	75%	88%	GDC Finance — project expenditure records; KfW disbursement reports	Administrative — quarterly financial reports	Quarterly	GDC Finance and M&E Sections
IN.2	KfW loan disbursement as % of EUR 80M facility	63%	80%	95%	KfW disbursement notices; GDC Finance records	Secondary — KfW quarterly disbursement notices	Quarterly	GDC Finance and M&E Sections
IN.3	Project technical staff deployed as % of planned establishment	82%	90%	95%	GDC HR — project team deployment records	Administrative — monthly HR deployment reports	Monthly — reported quarterly	GDC Project Manager and HR

4.2 Output Indicators

Code	Indicator	Baseline	Yr 1Target	Yr 2Target	Data Source	Collection Method	Frequency	Responsible
OP.1	Km of access roads constructed and commissioned	70km	70km maintained	70km maintained	GDC Infrastructure — road completion certificates; Implementation Consultant reports	Site inspection; contractor certificates	Quarterly	GDC Project Engineer

OP.2	Water supply system — % completion and daily delivery capacity	36%	80%	95%	GDC Infrastructure — completion certificates; water flow measurement records	Construction progress reports; flow measurement	Quarterly	GDC Project Engineer
OP.3	Number of drill pads constructed and commissioned — cumulative	11	15	19	GDC Infrastructure — pad completion certificates	Site inspection; contractor completion certificates	Quarterly	GDC Project Engineer
OP.4	Number of geothermal wells drilled — cumulative (production and reinjection)	20	26	30	GDC Drilling — well completion reports; QMC certificates; Implementation Consultant supervision	Well completion reports per well drilled	Per well — reported quarterly	GDC Drilling Manager
OP.5	Number of wells tested, logged and confirmed productive — cumulative	9	15	20	GDC Reservoir Engineering — well test reports; logging reports; Implementation Consultant	Well testing reports per well	Per well tested — reported quarterly	GDC Reservoir Engineer
OP.6	Geothermal steam capacity confirmed and available at wellhead — MW equivalent	85 MW	90 MW	100 MW	GDC Reservoir Engineering — well test reports; Feasibility Study; reservoir engineering modelling	Aggregated well test analysis; Feasibility Study	Quarterly — updated after each well test	GDC Reservoir Engineer
OP.7	Steam gathering system construction completion — %	0%	25%	70%	AMEA Power construction progress reports; GDC Project Engineer inspections; Implementation Consultant	Monthly construction progress reports; site inspection certificates	Monthly — reported quarterly	GDC Project Engineer and AMEA Power

4.3 Immediate Outcome Indicators

Code	Indicator	Baseline	Yr 1Target	Yr 2Target	Data Source	Collection Method	Frequency	Responsible
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IO.1	Steam supply agreement GDC–AMEA Power signed and operational (binary 0/1)	0 — not signed	1 — signed	1 — operational with delivery schedule	GDC Legal and Commercial — agreement documentation	Document review — signed and stamped agreement	Annual — upon milestone achievement	GDC Commercial and M&E Sections
IO.2	PPA signed between AMEA Power and KPLC — ratified by EPRA (binary 0/1)	0 — not ratified	PPA negotiations commenced	PPA signed — submitted to EPRA	EPRA regulatory gazette; KPLC commercial records; AMEA Power project records	Document review — PPA gazette notice and ratification certificate	Annual — upon milestone achievement	GDC Commercial and M&E Sections
IO.3	AMEA Power geothermal power plant construction commenced (binary 0/1)	0 — not commenced	FID made by AMEA Power	EPC contract or mobilised — construction commenced	AMEA Power construction progress reports; GDC Project Engineer site inspection; EPRA generation licence	Site inspection reports; construction commencement certificate	Quarterly	GDC Project Engineer and M&E Sections

4.4 Intermediate Outcome Indicators

Code	Indicator	Baseline	Yr 1 Target	Yr 2 Target	Data Source	Collection Method	Frequency	Responsible
MO.1	MW of geothermal power plant capacity developed and commissioned by AMEA Power at Paka	0 MW	0 MW — construction in progress	0 MW — construction advancing	AMEA Power commissioning certificate; EPRA generation licence gazette; KenGen grid synchronisation	Commissioning certificate; EPRA generation licence; grid synchronisation records	Annual — upon commissioning milestone	GDC Project Manager and M&E
MO.2	Annual electricity generated from Paka plant and supplied to national grid — GWh per year ($MW \times 0.90 \times 8,760 \div 1,000$)	0 GWh	0 GWh — under construction	0 GWh — under construction	AMEA Power generation records; KenGen grid offtake records; EPRA generation statistics	AMEA Power monthly generation reports; KPLC offtake records	Monthly — reported annually once operational	AMEA Power and GDC M&E
MO.3	Paka geothermal power plant capacity factor — % (target minimum 90% per charter design assumption)	0% — not operational	0% — under construction	0% — under construction	AMEA Power plant operations records; EPRA generation statistics; KenGen	AMEA Power monthly operations reports; EPRA dispatch data	Monthly — reported annually once operational	AMEA Power and GDC M&E

					dispatch records			
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4.5 Impact Indicators

Code	Indicator	Baseline	Yr 1 Target	Yr 2 Target	Data Source	Collection Method	Frequency	Responsible
IM.1	Cost of electricity generation from Paka project — USD cents per kWh — benchmarked against alternative generation sources (emergency: 22¢; oil-based: 16¢)	Paka projected: 6.5¢/kWh per charter	6.5¢/kWh confirmed in Feasibility Study	6.5¢/kWh confirmed in PPA tariff	GDC Commercial; Feasibility Study final report; AMEA Power PPA tariff; EPRA gazette	Feasibility Study report; GDC commercial cost analysis; AMEA Power PPA tariff submissions	Annual — upon FS completion, PPA agreement, and commissioning	GDC Commercial and M&E Sections
IM.2	Annual CO2 equivalent emissions avoided through Paka generation — tonnes CO2e (calculated: MW generated × 0.90 × 8,760 × 0.5 tCO2/MWh using EPRA grid emission factor)	0 tCO2e — not operational	0 tCO2e — under construction	0 tCO2e — under construction	AMEA Power generation records; EPRA published grid emission factor 0.5 tCO2/MWh	Calculated from MO.2 annual generation data using EPRA grid emission factor formula	Quarterly — reported annually once plant operational	GDC M&E Section
IM.3	Baringo Silali Paka Sub-Project MW contribution to Kenya installed geothermal capacity (national baseline: 943.7 MW per EPRA FY2023/24)	0 MW from Paka Sub-Project	0 MW — plant under construction	0 MW — plant approaching commissioning	AMEA Power commissioning certificate; EPRA Bi-Annual Energy and Petroleum Statistics Report	GDC commissioning records at plant commissioning; EPRA published capacity statistics	Annual — upon commissioning milestone	GDC M&E Section

4.6 Assumptions Monitoring

Code	Assumption	Risk if Assumption Fails	Likelihood	Impact	Monitoring Indicator
AS.1	Each well outputs minimum 5MW	Steam capacity falls below 100MW target	Medium	High	Average MW output per productive well from well testing reports
AS.2	90% well success rate achieved	More wells required — cost overrun and schedule delay	Medium	High	Ratio of productive wells to total wells drilled — cumulative

AS.3	AMEA Power secures adequate financing for power plant	Power plant delayed or cancelled	Medium	Very High	Financial Investment Decision milestone achieved — binary
AS.4	PPA tariff commercially viable for AMEA Power	AMEA Power unable to secure financing	Medium	Very High	PPA signed and EPRA ratified — binary milestone
AS.5	KETRACO transmission by April 2027	Plant commissioned but cannot evacuate power to grid	Medium	High	KETRACO construction percentage completion against charter schedule
AS.6	Steam quality meets AMEA Power plant specifications	Power plant design modification required	Low	High	Steam quality parameters — temperature, pressure, enthalpy at wellhead vs specifications
AS.7	No significant adverse geological events	Drilling programme disrupted — productive wells lost	Low	High	Geochemical monitoring results — quarterly reservoir pressure and fluid chemistry analysis
AS.8	Regulatory approvals obtained on schedule	Plant construction delayed	Low	Medium	Status of all required licences and permits against scheduled approval dates

5. Indicator Reference Sheet

The Indicator Reference Sheet provides full technical documentation for each result indicator — precisely what is being measured, how it is calculated, baseline and targets, responsible party, and data quality risks. The following presents the complete reference sheet for the three impact indicators — the most critical for donor accountability and the indicators that directly answer the question every reviewer asks of an M&E system: did the project make a difference? Reference sheets for all remaining 16 indicators (inputs, outputs, immediate and intermediate outcomes) follow the identical field structure shown below and are available in full in Annex B.

OP.6 — Output	
Indicator name	Geothermal steam capacity confirmed and available at wellhead — MW equivalent
Definition	Total MW equivalent of geothermal steam capacity confirmed productive at wellhead and available for delivery to IPP power plant — based on aggregated well testing results and reservoir engineering assessment — the culminating output that bridges drilling activities to power generation outcomes.
Unit of measurement	Megawatts equivalent (MWe)
Baseline value	85MW equivalent confirmed at wellhead per charter
Targets	• Year 1: 90MW • Year 2: 100MW • Final: 100MW — full design capacity confirmed and available
Data source	GDC Reservoir Engineering Department — well test reports; Feasibility Study final report
Collection method	Aggregated well test analysis; reservoir engineering assessment; Feasibility Study findings
Collection frequency	Quarterly — updated after each well test
Responsible party	GDC Reservoir Engineer and M&E Section
Users	KfW, GRMF, AMEA Power, GDC Board
Use of information	KfW resource development milestone; AMEA Power investment trigger for power plant construction decision; GDC Board commercial viability confirmation
SDG crosswalk	SDG 7.2.1 — Renewable energy share; SDG 7.b.1 — Installed renewable energy generating capacity
Disaggregation	By well — individual well MW contribution; by field zone
Contribution vs attribution	Attribution — GDC directly responsible for steam field development and wellhead capacity
Data quality risks	Wellhead capacity may not be sustained at stated levels due to reservoir pressure decline or fluid chemistry changes — point-in-time measurements may overstate sustained capacity
Mitigation	Use sustained wellhead flow test data rather than instantaneous peak measurements — conduct minimum 72-hour flow tests per well; conduct periodic reservoir pressure monitoring

IM.1 — Impact	
Indicator name	Cost of electricity generation from Paka geothermal project — USD cents per kWh — benchmarked against alternative generation sources
Definition	Unit cost of electricity generation from Paka geothermal project in USD cents per kWh — benchmarked against the cost of alternative generation sources currently used in Kenya. Emergency power: 22 US cents per kWh; Oil-based power: 16 US cents per kWh; Projected Paka geothermal: 6.5 US cents per kWh — per project charter.
Unit of measurement	USD cents per kWh
Baseline value	Emergency power: 22¢/kWh; Oil-based: 16¢/kWh; Paka projected: 6.5¢/kWh (per charter)
Targets	• Year 1: 6.5¢/kWh confirmed in Feasibility Study final report • Year 2: 6.5¢/kWh confirmed in PPA tariff agreed between AMEA Power and KPLC • Final: 6.5¢/kWh or below — confirmed at commissioning and maintained through first full year of commercial operation
Data source	GDC Commercial Department — steam tariff and power cost analysis; Feasibility Study final report; AMEA Power tariff submission to EPRA
Collection method	Feasibility Study report; GDC commercial cost analysis; AMEA Power PPA tariff submissions; EPRA gazette notices
Collection frequency	Annual — upon Feasibility Study completion, PPA agreement, and at commissioning
Responsible party	GDC Commercial and M&E Sections
Users	KfW, GDC Board, Ministry of Energy, EPRA, Government of Kenya
Use of information	KfW cost competitiveness evidence for development finance portfolio; GDC Board commercial viability confirmation; Ministry of Energy tariff policy evidence; EPRA tariff setting reference
SDG crosswalk	SDG 7.1.1 — Affordable, reliable and modern energy; SDG 7.3.1 — Energy intensity improvement
Disaggregation	By cost component — steam cost, power plant O&M, transmission, financing
Contribution vs attribution	Attribution — cost of Paka geothermal power is directly calculable from project cost data and Feasibility Study
Data quality risks	Projected costs may diverge from actual costs due to construction cost overruns, financing cost changes, or O&M cost variations
Mitigation	Update cost estimate at each project milestone — Feasibility Study, FID, commissioning, and first year of operation — and report variance from charter projection

IM.2 — Impact	
Indicator name	Annual CO2 equivalent emissions avoided through Paka geothermal generation — tonnes CO2e
Definition	Tonnes of CO2 equivalent emissions avoided annually as a result of Paka geothermal electricity generation displacing fossil fuel thermal generation in Kenya's national grid — calculated using EPRA published national grid emission factor of 0.5 tCO2/MWh.

Unit of measurement	Tonnes CO2 equivalent per year (tCO2e/year)
Baseline value	0 tCO2e — plant not yet operational
Targets	• Year 1: 0 tCO2e — plant under construction • Year 3: 0 tCO2e — plant under construction • Final: 277,000 to 394,200 tCO2e avoided annually — based on 70 to 100MW at 90% capacity factor using EPRA grid emission factor ($\text{MW} \times 0.90 \times 8,760 \times 0.5 \text{ tCO}_2/\text{MWh}$)
Data source	AMEA Power generation records via MO.2; EPRA published grid emission factor — 0.5 tCO2/MWh per EPRA Statistics Report FY 2023/2024
Collection method	Calculated from MO.2 annual generation data using EPRA grid emission factor — formula: $\text{GWh generated} \times 1,000 \times 0.5 \text{ tCO}_2/\text{MWh}$
Collection frequency	Quarterly — reported annually once plant operational
Responsible party	GDC M&E Section
Users	KfW, GRMF, NEMA, Ministry of Environment, GDC Board, Green Climate Fund
Use of information	KfW climate finance portfolio reporting; GRMF climate impact evidence; NEMA environmental compliance; Ministry of Environment Kenya NDC UNFCCC reporting; GDC Board sustainability reporting
SDG crosswalk	SDG 13.2.2 — Total greenhouse gas emissions per year; SDG 7.2.1 — Renewable energy share in total final energy consumption
Disaggregation	By project phase — 70MW scenario vs 100MW scenario; annual vs cumulative lifetime emissions avoided
Contribution vs attribution	Attribution — emissions avoided calculation is directly and exclusively based on Paka plant generation data using standardised internationally recognised methodology
Data quality risks	EPRA grid emission factor is subject to periodic revision — changes affect comparability over time. Capacity factor assumption may vary from actual operational performance
Mitigation	Always use most recently published EPRA grid emission factor — document factor used in each annual report. Note any factor revisions and restate prior year figures for comparability

IM.3 — Impact

Indicator name	Baringo Silali Paka Sub-Project MW contribution to Kenya installed geothermal capacity
Definition	MW of geothermal generation capacity from Paka Sub-Project commissioned and operational — tracked as GDC's specific contribution to Kenya's national installed geothermal capacity. National baseline: 943.7MW installed geothermal per EPRA FY 2023/2024.
Unit of measurement	Megawatts (MW) — Paka contribution tracked against national baseline
Baseline value	0MW from Paka Sub-Project — national installed geothermal baseline: 943.7MW per EPRA FY 2023/2024
Targets	• Year 1: 0MW — plant under construction; national baseline tracked

	• Year 2: 0MW — plant approaching commissioning; national baseline tracked • Final: 70 to 100MW commissioned from Paka — contributing to national total of 1,013 to 1,043.7MW installed geothermal capacity
Data source	AMEA Power commissioning certificate; EPRA Bi-Annual Energy and Petroleum Statistics Report for national total
Collection method	GDC commissioning records at plant commissioning; EPRA published capacity statistics for national total verification
Collection frequency	Annual — upon commissioning milestone; national figure updated bi-annually per EPRA reporting cycle
Responsible party	GDC M&E Section
Users	KfW, GDC Board, Ministry of Energy, KenGen, EPRA
Use of information	KfW ultimate project impact evidence; GDC Board mandate delivery accountability — 1,065MW by 2030 target; Ministry of Energy national capacity planning and IRENA renewable energy reporting; KenGen generation scheduling
SDG crosswalk	SDG 7.2.1 — Renewable energy share in total final energy consumption; SDG 7.b.1 — Installed renewable energy generating capacity in developing countries
Disaggregation	Paka Sub-Project contribution tracked separately from national total; Paka contribution as percentage of 1,065MW GDC mandate target
Contribution vs attribution	Attribution for Paka contribution — MW commissioned from Paka is directly and solely attributable to GDC project delivery. National total is contextual — contribution framing applies
Data quality risks	Commissioning milestones may be delayed by technical, geological, procurement, or regulatory factors — affecting target achievement timing
Mitigation	Distinguish between target achievement delays caused by project implementation factors versus external factors — document root cause of any milestone delay in annual reports

Note: Full indicator reference sheets for all 19 result indicators — IN.1 through IM.3 — follow this identical structure and are provided in Annex B of this document.

6. M&E Plan

6.1 Introduction and Purpose

This M&E Plan operationalises the M&E system designed for the Paka-Baringo Silali project. It translates the results framework, Theory of Change, and indicator reference sheet into a practical, implementable system serving four functions: accountability to KfW, GRMF, and development partners; management decision-making for GDC project teams; learning for replication of GDC’s PPP model; and transparency to communities and the Kenyan public.

6.2 Data Collection Methodology

The M&E system uses a mixed methods approach combining quantitative performance tracking with qualitative depth inquiry. Quantitative methods include: administrative data from GDC project records; secondary data from EPRA published statistics and KenGen/KPLC records; structured household surveys (Tool 1); and structured observation (Tool 2). Qualitative methods include: key informant interviews (Tool 4) and document review. Every significant finding is triangulated across at least two independent data sources before being reported as a conclusion.

6.3 Data Flow Architecture

The data flow operates through four stages:

1. Data Generation — from GDC Drilling, Infrastructure, Finance, HR, and Environmental sections; AMEA Power; community households; EPRA published reports; and KenGen/KPLC records.
2. Data Submission and Aggregation — GDC Section Heads submit monthly data to M&E Section by the 5th of each month. M&E Section aggregates into the master monitoring matrix by the 15th of each month.
3. Data Quality Assurance — four-step process: completeness check, validity check, triangulation, and trend consistency check. Formal annual Data Quality Assessment conducted.
4. Analysis, Reporting, and Use — verified data flows into the Tableau Monitoring Dashboard (monthly update), periodic progress reports, and bi-annual evaluation reviews.

6.4 Data Quality Assurance

The M&E system applies the seven internationally recognised dimensions of data quality: validity, reliability, timeliness, precision, integrity, confidentiality, and accessibility. A formal Data Quality Assessment is conducted annually on the top five priority indicators. Community survey instruments are pre-tested with 10–15 households before each deployment round. All enumerators attend a half-day training session before each survey round.

6.5 Reporting Schedule and Formats

Report Type	Frequenc y	Due Date	Audience	Content Summary
Monthly Dashboard Update	Monthly	20th of following month	GDC Project Manager, Senior Management, Section Heads	Updated indicator values; assumption status; ESCP summary; drilling status; financial utilisation.

Monthly Internal M&E Report	Monthly	25th of following month	GDC Project Manager, GDC Board (selected)	Performance summary; financial utilisation; assumption monitoring; issues and actions; decisions required.
Quarterly Donor Progress Report	Quarterly	30 days after quarter end	KfW, GRMF, Ministry of Finance	Full results chain indicators; financial utilisation; assumption monitoring; ESCP compliance; risk register; next quarter work plan.
Bi-Annual Community Engagement Report	Bi-annual	45 days after survey	Community leaders, Baringo County Government, KfW	Community survey findings; KII findings from community leaders; trend comparison; concerns and GDC response plan.
Annual Results Report	Annual	31st March	KfW, GRMF, Ministry of Energy, GDC Board, AMEA Power	Full results chain performance; year-on-year trends; assumption findings; ESCP annual compliance; lessons learned; updated risk register; next year work plan.
Ad Hoc Issue Reports	As required	Within 5 working days	KfW, GDC Board, MoE	Issue description; impact assessment; management response; timeline for resolution.

6.6 M&E Budget

M&E Activity	Annual Cost USD	Notes
GDC M&E Section staffing — 2 FTE M&E officers	80,000	Salaries and benefits
Community perception survey — bi-annual	30,000	Enumerator training, field costs, data entry, analysis
Environmental safeguards monitoring — Environmental Consultant	60,000	Monthly site visits, reporting
Data systems and Tableau licence	5,000	Software and platform costs
Data Quality Assessment — annual	15,000	External DQA facilitator
Key informant interviews — bi-annual	8,000	Travel, facilitation, transcription
Annual results report production	10,000	Design, printing, translation
M&E training and capacity building	12,000	Staff training, tools updates

Monitoring visits and site inspections	20,000	GDC M&E Section field visits
Contingency — 10%	24,000	Unforeseen M&E costs
TOTAL ANNUAL M&E BUDGET	USD 264,000	~0.13% of total project budget

6.7 RACI Matrix — Roles and Responsibilities

M&E Function	GDC M&E	Proje ct Mgr	Section Heads	Enviro n. Officer	Enviro n. Consu lt.	AMEA Power	KfW	Com m. Rela tions
M&E system design and updates	R/A	C	C	C	C	I	I	I
Monthly performance data — Tool 3	A	I	R	R	I	R	I	I
Community survey design — Tool 1	R/A	C	I	C	C	I	C	C
Community survey implementation	R	I	I	I	I	I	I	C
ESCP checklist — Tool 2	A	I	I	R	R	I	I	I
Key informant interviews — Tool 4	R/A	C	I	I	I	C	C	C
Monthly dashboard update	R/A	I	I	I	I	I	I	I
Quarterly donor progress report	R	A	C	C	C	C	I	I
Annual results report	R	A	C	C	C	C	I	C
Data Quality Assessment	R/A	C	C	I	C	I	I	I
Assumption monitoring	R	A	C	C	I	C	I	I
KfW supervision mission support	R	A	C	C	C	C	R	I

Key: R = Responsible, A = Accountable, C = Consulted, I = Informed

6.8 M&E Capacity and Staffing

The M&E system requires three dedicated staff positions: (1) M&E Specialist — Head of M&E Section — responsible for overall M&E system management, donor reporting, annual results report, and DQA oversight; requires MSc Statistics or Social Sciences and 5+ years development sector M&E experience; (2) M&E Officer — Data Systems — responsible for monthly data aggregation, quality assurance, Tableau dashboard management, and community survey analysis; requires BSc Statistics or Data Science and Tableau proficiency; (3) Community Relations Officer — M&E Support — responsible for community survey

coordination, KII facilitation, and community engagement reporting; requires local language proficiency and community engagement experience.

6.9 Technology Platform and Data Management

Primary data management platform: Microsoft Excel — structured workbook with version control.

Dashboard and visualisation platform: Tableau — dashboard published on Tableau Public for stakeholder access. Data management standards: World Bank DIME Data Handbook — reproducibility, transparency, and version control principles applied throughout (note: this references the DIME data management methodology only; the project itself is financed by KfW, GRMF, and the Government of Kenya — see Section 2.4).

All M&E data files follow a standardised naming convention:

GDC_PBSGP_[DataType]_[YYYYMM]_v[VersionNumber]. The master dataset is backed up to GDC's secure server daily and to an offsite location weekly. All M&E data is retained for a minimum of 10 years after project closure in accordance with KfW data retention requirements for funded projects.

6.10 Adaptive Management and Learning Framework

Monthly management reviews are convened by the GDC Project Manager on the 1st working day of each month using the Tableau dashboard and monthly M&E report as primary agenda inputs — reviewing performance against key output indicators, assumption status, ESCP compliance issues, financial utilisation variance, and decisions required.

Bi-annual learning reviews go beyond performance tracking to ask: what is working better than expected and why; what is not working and why; what should we do differently in the next six months; what have we learned that is relevant to GDC's other geothermal projects.

Three specific trigger points require a formal written management response submitted to KfW: (1) cumulative well success rate falls below 80% — triggers reservoir engineering review and revised drilling plan; (2) AMEA Power Financial Investment Decision not achieved within 6 months of PPA ratification — triggers commercial review meeting with GDC, AMEA Power, and KfW; (3) more than 5 community grievances pending resolution beyond agreed timeframe — triggers community relations review and stakeholder engagement plan revision.

7. Data Collection Tools

Four purpose-designed data collection tools operationalise the M&E system. Each tool is grounded in the monitoring matrix and designed for the specific data collection context of the Paka-Baringo Silali project. Full instrument texts including all questions, response options, and administration protocols are provided in Annexes E through H.

Tool	Purpose	Respondents	Indicators	Frequency	Administered by
Tool 1 — Community Perception and Welfare Survey	Measure community attitudes, benefit-sharing satisfaction, livelihood changes, water access, and employment perceptions.	Household heads and adult members — minimum 200 HH, stratified by location, gender, livelihood — within 15km of Paka field.	Community welfare; AS.7, AS.8	Bi-annual	GDC M&E Officer and Community Relations Officer
Tool 2 — Environmental and Social Safeguards Checklist	Track compliance with the Environmental and Social Commitment Plan (ESCP) and KfW environmental and social safeguards requirements across 8 safeguard domains.	GDC Environmental Officer and Environmental and Social Management Consultant.	Safeguards compliance; community grievances	Monthly	GDC Environmental Officer and Environmental Consultant
Tool 3 — Project Performance Tracking Template	Monthly output and input tracking — drilling progress, infrastructure milestones, financial utilisation, commercial milestone status.	GDC Section Heads and AMEA Power Project Team.	OP.1–OP.7, IN.1–IN.3, IO milestones, assumptions status	Monthly — due 5th of following month	GDC M&E Section in consultation with Section Heads
Tool 4 — Key Informant Interview Guide	Qualitative depth on outcomes, commercial progress, assumption status, and stakeholder perspectives.	AMEA Power, EPRA, KfW, KPLC, community leaders, MoEP officials, KETRACO.	IO.1, IO.2, IO.3, MO.1–MO.3; qualitative assumption monitoring	Bi-annual	GDC M&E Officer or designated evaluator

7.1 Tool 1 — Community Perception and Welfare Survey

Administration: Face-to-face interview — enumerator administered in Pokot, Tugen, or Swahili. Sample size: minimum 200 households. Duration: 35 to 45 minutes. The full survey instrument (56 questions across 7 sections) is provided in Annex E.

- Section A — Household identification and profile (gender, age, education, livelihood, household composition, employment by GDC)
- Section B — Project awareness and knowledge (sources of information, level of understanding, consultation history)

- Section C — Water access (primary source, change since project commencement, satisfaction with GDC water supply, livestock water access)
- Section D — Infrastructure and access (road benefits, types of benefit, land or property damage)
- Section E — Employment and economic benefits (employment opportunities created, local priority, business benefits, household economic impact)
- Section F — Community relations and grievances (relationship rating, grievance awareness and resolution, information transparency, benefit fairness)
- Section G — Overall perception and future expectations (overall sentiment, long-term development expectation, open-ended priorities and comments)

7.2 Tool 2 — Environmental and Social Safeguards Monitoring Checklist

Frequency: Monthly. The full checklist (47 items across 8 sections) covers: air quality and noise; water resources and groundwater protection; land and soil management; biodiversity and ecosystem; occupational health and safety; community health, safety and security; social and economic safeguards; and ESCP compliance summary. Full instrument in Annex F.

7.3 Tool 3 — Project Performance Tracking Template

Frequency: Monthly — due by the 10th of the following month. The template covers seven sections: financial performance; human resources; infrastructure output tracking; drilling output tracking; commercial milestones; risk and assumptions monitoring; and issues and actions. Full template in Annex G.

7.4 Tool 4 — Key Informant Interview Guide

Duration: 45 to 60 minutes. **Frequency:** Bi-annual. The guide covers seven sections — introduction script, informant profile, project progress and performance, commercial and regulatory enabling conditions, community and social outcomes, energy security and impact perspectives, and learning and recommendations. Full guide in Annex H.

8. Evaluation Plan

8.1 Purpose and Rationale

Monitoring tells us whether the project is doing what it said it would do. Evaluation tells us whether it is making a difference — and why. The evaluation plan serves three specific purposes: accountability to KfW and GRMF; adaptive management at strategic decision points — the AMEA Power FID, the PPA negotiation outcome, the transition to power generation; and learning for GDC’s broader mandate, since the Paka project is a test case for GDC’s PPP development model.

8.2 Evaluation Criteria — OECD DAC

Criterion	Evaluation Question for This Project
Relevance	Is the project addressing the right problem in the right way — and does it remain aligned with Kenya’s evolving energy policy context?
Effectiveness	Is the project achieving its intended outcomes — commercial agreements, power plant development, grid supply?
Efficiency	Are the project’s results being achieved at an appropriate cost and within planned timeframes?
Impact	Is the project contributing to the intended long-term changes in electricity costs, emissions, energy security, and community welfare?
Sustainability	Will the project’s results endure beyond the project period — is the geothermal resource sustainable, is the commercial model viable long-term?
Coherence	Is the project well coordinated with Kenya’s broader energy sector programmes — KETRACO transmission, KPLC distribution, government electrification targets?

8.3 Four Evaluation Questions

EQ1 — Relevance and Coherence	
Evaluation question	To what extent does the Paka-Baringo Silali project remain aligned with Kenya’s national energy priorities, donor strategic frameworks, and the evolving needs of project-adjacent communities — and is it well coordinated with complementary sector investments?
OECD DAC criteria	Relevance, Coherence
Indicators informing	IM.1, IM.3, IO.2, AS.5 — plus document review of LCPDP, energy policy, and community needs assessment

EQ2 — Effectiveness	
Evaluation question	Is the project achieving its intended immediate and intermediate outcomes — and what factors are enabling or constraining outcome achievement?
OECD DAC criteria	Effectiveness
Indicators informing	IO.1, IO.2, IO.3, MO.1, MO.2, MO.3 — plus Tool 1 community survey and Tool 4 KII findings

EQ3 — Efficiency and Value for Money	
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Evaluation question	Are the project's results being achieved at an appropriate cost and within planned timeframes — and does the project represent good value for money for development finance invested?
OECD DAC criteria	Efficiency
Indicators informing	IN.1, IN.2, OP.4, OP.6, IM.1, IM.2 — plus financial records and cost benchmarking

EQ4 — Sustainability and Impact Contribution	
Evaluation question	Are the project's intended long-term impacts materialising — and are the conditions in place for these impacts to be sustained beyond the project period?
OECD DAC criteria	Impact, Sustainability
Indicators informing	IM.1, IM.2, IM.3, MO.2, MO.3 — plus qualitative evidence from KIIs and community surveys

8.4 Evaluation Schedule

Evaluation	Type	Timing	Primary Criteria	Evaluator	Budget USD
Mid-term review	Formative — internal	Year 2 of project	Relevance, Effectiveness, Efficiency, Coherence	GDC M&E Specialist + KfW project officer	40,000–60,000
Pre-commissioning evaluation	Summative — steam field phase	Years 4–5	Effectiveness, Efficiency, Coherence, Sustainability	Independent external evaluator	80,000–120,000
Post-commissioning impact evaluation	Summative — impact	Years 7–8	Impact, Sustainability, Effectiveness, Efficiency	Independent external evaluator team	120,000–180,000
TOTAL EVALUATION BUDGET					240,000–360,000

Three evaluations are conducted across the full project lifecycle (2025–2032) — note this differs from the 24-month illustrative dataset used for the dashboard in Section 9, which shows only the early steam field development phase.

8.5 Post-Commissioning Impact Evaluation Design

Contribution Analysis — primary method: Builds a structured causal argument tracing the project's results chain from activities through to impact indicators, testing evidence at each link. The contribution story for Paka runs: GDC drills and confirms wells → steam capacity confirmed at commercially viable cost → AMEA Power commissions power plant → electricity generated at 6.5 US cents per kWh → thermal generation displaced → electricity costs reduced → CO2 avoided annually.

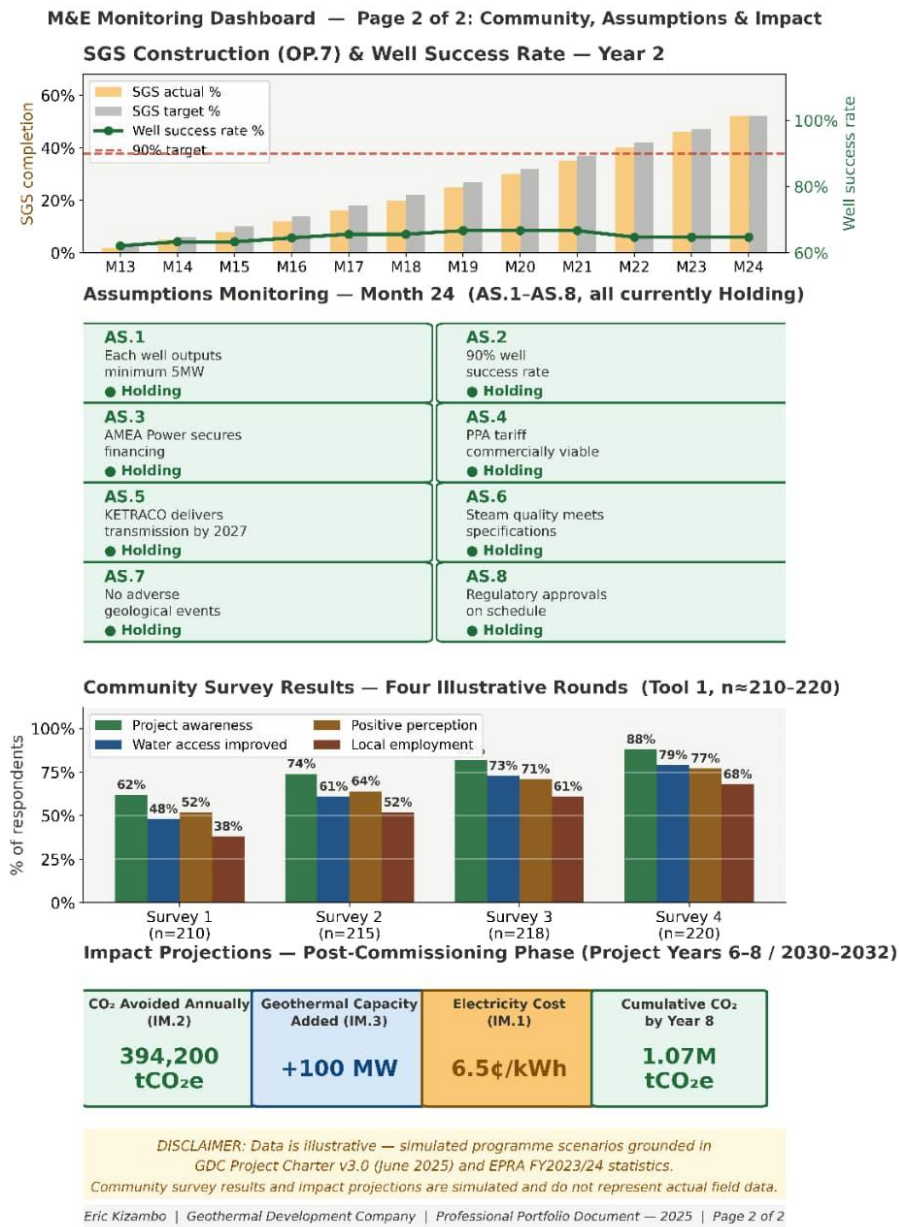
Difference-in-Differences Analysis — supplementary method: For community welfare impacts, compares Baringo county welfare indicators before and after plant commissioning against comparable county indicators over the same period, using Kenya National Bureau of Statistics household surveys supplemented by GDC's bi-annual community perception surveys (Tool 1).

8.6 Evaluation Management and Quality Assurance

The mid-term review is internally led by the GDC M&E Specialist. The pre-commissioning and post-commissioning evaluations are externally commissioned to ensure independence. Quality assurance includes: Terms of Reference reviewed and approved by KfW before evaluator recruitment; evaluator selection criteria covering relevant energy sector experience and independence; inception report reviewed before fieldwork; draft report reviewed for factual corrections only; and final report published within 30 days of acceptance, accompanied by a management response.

8.7 Learning and Knowledge Management

Evaluation findings are disseminated through stakeholder workshops within 60 days of publication; two-page internal learning briefs for all GDC Section Heads; submission to AfrEA (African Evaluation Association) conference and open-access development journals; and sharing with GRMF (African Development Bank) for portfolio-level learning on geothermal risk mitigation across Africa.



9.1 Dashboard Structure

The dashboard is organised across panels covering the complete M&E story from inputs through to impact projections:

- Executive KPIs panel: Four headline indicators — steam capacity, wells drilled, budget utilised, SGS construction — with actual vs target status.
- Output milestone tracker: All output milestones with achievement status — Achieved, On Track, Delayed, or Pending.
- Wells drilled vs confirmed productive chart: Time series showing cumulative wells drilled and confirmed against targets across the illustrative window.
- Steam capacity at wellhead chart: Growth from baseline to 100MW achievement.
- Financial performance chart: Budget utilisation percentage and KfW disbursement percentage against targets.

- SGS construction progress chart: Monthly progress against target.
- Well success rate chart: Actual cumulative well success rate against the 90% target assumption (AS.2).
- Assumptions monitoring panel: Current status of all eight assumptions.
- Community survey results panel: Four illustrative survey rounds showing awareness, water access, perception, and employment.
- Impact projections panel: Post-commissioning projections — CO2 avoided, geothermal capacity added, electricity generation cost, cumulative CO2 avoided.

9.2 Data Sources and Simulated Dataset

The dashboard draws from the simulated dataset structured across eight Excel worksheets: Monthly_Data, Target_vs_Actual, Drilling_Progress, Financial_Tracking, Community_Survey, Impact_Projections, Assumptions_Tracker, and Metadata — each grounded in the GDC Project Charter Version 3.0 (June 2025) baseline figures and EPRA FY2023/2024 published statistics.

Annexes

The following annexes form part of this M&E System Design document:

- Annex A** Stakeholder Matrix — Full Analysis Table with Roles, Interests, Influence, M&E Data Needs, and Reporting Frequencies for All 13 Stakeholders
- Annex B** Full Indicator Reference Sheet — Complete Technical Documentation for All 19 Result Indicators (IN.1 through IM.3)
- Annex C** Data Flow Architecture Diagram — Four-Stage Data Flow from Generation through Quality Assurance to Reporting
- Annex D** RACI Matrix — Full Responsibilities Table for All 14 M&E Functions Across 8 Stakeholder Groups
- Annex E** Tool 1 — Community Perception and Welfare Survey — Full Instrument (56 Questions, 7 Sections, Administration Protocol)
- Annex F** Tool 2 — Environmental and Social Safeguards Monitoring Checklist — Full Instrument (47 Items, 8 Sections, Corrective Action Plan Template)
- Annex G** Tool 3 — Project Performance Tracking Template — Full Monthly Template (7 Sections, All Indicator Tables)
- Annex H** Tool 4 — Key Informant Interview Guide — Full Guide (7 Sections, 4 Informant Categories, Introduction and Closing Scripts)
- Annex I** SDG Crosswalk Table — All 19 Indicators Mapped to Specific SDG Targets and Global Indicator Framework References
- Annex J** M&E Budget — Detailed Annual Cost Breakdown by Activity with Notes on Cost Assumptions
- Annex K** M&E Monitoring Dashboard — Full Resolution Image and Simulated Dataset (GDC_PBS_ME_Simulated_Dataset_v1.0.xlsx)
- Annex L** Theory of Change Diagram — Full Resolution Image — Paka-Baringo Silali 100MW Geothermal Steam Field Development — Version 1.0 — 2025

End of M&E System Design Document

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